

test 1 — same $\beta/C = 0.02$

$$\beta = 1, C = 50$$

$$G = 2.281540715429e+02$$

$$\beta = 2, C = 100$$

$$G = 2.281540715429e+02$$

$$\beta = 4, C = 200$$

$$G = 2.281540715429e+02$$

identical to 13 digits (spread 0.00e+00)

**a manifest symmetry of the integrand — a regression check,
not evidence that β/C is a real knob**

test 2 — different β/C

$$\beta/C = 0.02$$

$$G = 2.281540715429e+02$$

$$\beta/C = 0.01$$

$$G = 3.226723694195e+02$$

$$\beta/C = 0.005$$

$$G = 4.563373910491e+02$$

G changes by 41% — the ratio genuinely matters